

The Random Purification Channel

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May 29, 2026

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Conventions

- A complex, finite dimensional, inner product space (Hilbert space) is denoted by $\mathcal{H}$.
- We denote elements of $\mathcal{H}$, or *kets*, as $|\cdot\rangle$. The elements of the dual space $\mathcal{H}^*$, or *bras*, are denoted as $\langle \cdot |$. In particular, $\langle \cdot |$ is the adjoint of $|\cdot\rangle$.
- We denote the standard basis on $\mathcal{H}$ as $\{|i\rangle\}$, with $|i\rangle$ being the i th basis vector.
- We denote the space of linear operators between $\mathcal{H}_A$ and $\mathcal{H}_B$ as $L(\mathcal{H}_A, \mathcal{H}_B)$, and we denote space of square operators on $\mathcal{H}_A$ as $L(\mathcal{H}_A)$.
- The inner product on $L(\mathcal{H})$ is the *Hilbert-Schmidt* inner product, given by:

$$\langle A, B \rangle_{HS} = \text{Tr}(A^\dagger B)$$

for $A, B \in L(\mathcal{H})$.

Mathematical Preliminaries

Definition

Let $\mathcal{H}$ be a Hilbert space. A quantum state is some $\rho \in L(\mathcal{H})$ such that

1. $\langle \psi | \rho | \psi \rangle \geq 0$ for all $\psi \in \mathcal{H}$
2. $\text{Tr } \rho = 1$

The space of states is denoted $S(\mathcal{H})$.

Definition

A state ρ is called *pure* if $\text{rank}(\rho) = 1$ and it is called *mixed* otherwise.

If $\rho \in S(\mathcal{H})$ is pure, then there exists some $|\psi\rangle \in \mathcal{H}$ such that $\rho = |\psi\rangle\langle\psi|$. If ρ is a mixed state then it can be written as:

$$\rho = \sum_{i=1}^n p_i |\psi_i\rangle\langle\psi_i|$$

where $\{p_i\}_{i=1}^n$ are eigenvalues of ρ and $\{|\psi_i\rangle\}_{i=1}^n$ is a corresponding basis of orthonormal eigenvectors.

Quantum Channels

Definition (Quantum Channel)

A quantum channel is some linear $\mathcal{E} : L(\mathcal{H}_A) \rightarrow L(\mathcal{H}_B)$ such that for all ρ

1. Trace-preserving (TP): $\text{Tr}(\mathcal{E}(\rho)) = \text{Tr}(\rho)$
2. Completely positive (CP): If the map $\mathcal{E} \otimes I_{\mathcal{K}} : L(\mathcal{H}_A \otimes \mathcal{K}) \rightarrow L(\mathcal{H}_B \otimes \mathcal{K})$ is positive for all ancillary spaces $\mathcal{K}$.

Quantum channels are also referred to as CPTP maps.

Theorem

Let $\mathcal{E} : L(\mathcal{H}_A) \rightarrow L(\mathcal{H}_B)$ be a linear map. Then the following are equivalent:

1. $\mathcal{E}$ is completely positive.
2. There exists $\{F_i\} \subseteq L(\mathcal{H}_A, \mathcal{H}_B)$ such that $\mathcal{E}(X) = \sum_i F_i X F_i^\dagger$.

Furthermore, if $\mathcal{E}$ is trace preserving if and only if $\sum_i F_i^\dagger F_i = I$.

Symmetric Subspace

Definition (Action of Symmetric Group)

Let S_n be the symmetric group on n elements and let $\mathcal{H}$ be a d -dimensional Hilbert space. Then for all $\pi \in S_n$, we define $U_\pi : \mathcal{H}^{\otimes n} \rightarrow \mathcal{H}^{\otimes n}$ to be the unitary such that

$$U_\pi(|i_1\rangle \otimes \cdots \otimes |i_n\rangle) = |i_{\pi^{-1}(1)}\rangle \otimes \cdots \otimes |i_{\pi^{-1}(n)}\rangle \quad (1)$$

for all $i_1, \dots, i_n \in [d]$.

Example

Let $d = 2, n = 3$ with $\pi = (12)(3)$.

$$U_{(12)(3)}(|2\rangle \otimes |1\rangle \otimes |2\rangle) = |1\rangle \otimes |2\rangle \otimes |2\rangle \quad (2)$$

Symmetric Subspace

We call $\{X \in L(\mathcal{H}^{\otimes n}) : U_\pi X U_\pi^\dagger = X \text{ for all } \pi \in S_n\}$ the symmetric subspace of $L(\mathcal{H}^{\otimes n})$.

Definition (Symmetrizer)

We define $\text{Sym}_n : L(\mathcal{H}^{\otimes n}) \rightarrow L(\mathcal{H}^{\otimes n})$ such that

$$\text{Sym}_n(X) = \frac{1}{n!} \sum_{\pi \in S_n} U_\pi X U_\pi^\dagger \quad (3)$$

Lemma

Sym_n is a CPTP map.

Proof.

Kraus operators. □

Symmetric Subspace

Lemma

Sym_n is the orthogonal projection onto the symmetric subspace of $L(\mathcal{H}^{\otimes n})$ w.r.t the Hilbert-Schmidt inner product.

Lemma

Let $\dim \mathcal{H} = d$. Then, $\dim \text{Sym}_n(L(\mathcal{H}^{\otimes n})) = \binom{d^2+n-1}{n}$.

Proof of Dimension

1. Let $\{E_i\}_{i=1}^{d^2}$ be an orthonormal basis for $L(\mathcal{H})$. Then $\{E_{i_1} \otimes \cdots \otimes E_{i_n}\}$ for $i_1, \dots, i_n \in [d^2]$ is an orthonormal basis for $L(\mathcal{H}^{\otimes n})$.
2. We will symmetrize each basis vector and count the number which remain linearly independent.
3. Observe that $\text{Sym}_n(U_\sigma X U_\sigma^\dagger) = \text{Sym}$ for all $\sigma \in S_n$.

$$\text{Sym}_n(U_\sigma X U_\sigma^\dagger) = \sum_{\pi \in S_n} U_\sigma U_\pi X U_\pi^\dagger U_\sigma^\dagger = \sum_{\pi \in S_n} U_\sigma U_\pi X (U_\sigma U_\pi)^\dagger = \sum_{\pi' \in S_n} U_{\pi'} X U_{\pi'}^\dagger \quad (4)$$

4. Observe that two basis operators Φ_1, Φ_2 have the same multiplicities if and only if they differ by a permutation U_σ and so
 $\Phi_1 = U_\sigma \Phi_2 U_\sigma^\dagger \implies \text{Sym}_n(\Phi_1) = \text{Sym}_n(U_\sigma \Phi_2 U_\sigma^\dagger) = \text{Sym}_n(\Phi_2)$

Example

Let $\Phi_1 = E_1 \otimes E_2 \otimes E_1 \otimes E_2$ and $\Phi_2 = E_1 \otimes E_1 \otimes E_2 \otimes E_2$. Then,

$$U_{(1)(23)(4)}(\Phi_1)U_{(1)(23)(4)}^\dagger = \Phi_2 \quad (5)$$

Dimension Proof Continued

1. We will show that if basis vectors Φ_1, Φ_2 have different multiplicities then $\text{Sym}_n(\Phi_1) \perp \text{Sym}_n(\Phi_2)$.

$$\langle \text{Sym}_n(\Phi_1), \text{Sym}_n(\Phi_2) \rangle_{HS} = \langle \Phi_1, \text{Sym}_n(\Phi_2) \rangle_{HS} = \frac{1}{n!} \sum_{\pi \in S_n} \langle \Phi_1, U_\pi \Phi_2 U_\pi^\dagger \rangle_{HS} = 0 \quad (6)$$

2. The number of linearly independent basis vectors after symetrization = # of different multiplicity types = # of ways to place n identical balls in d^2 distinct bins
 $= \binom{n+d^2-1}{n}$

Partial Trace & Purification

Definition (Partial Trace)

Let $\mathcal{H}_A \otimes \mathcal{H}_B$ be a tensor product space. The *partial trace* with respect to B is the unique linear map $\text{Tr}_B : L(\mathcal{H}_A \otimes \mathcal{H}_B) \rightarrow L(\mathcal{H}_A)$ such that:

$$\text{Tr}_B(M \otimes N) = \text{Tr}(N)M$$

for all elementary tensors $M \otimes N \in L(\mathcal{H}_A) \otimes L(\mathcal{H}_B)$.

Definition (Purification)

Let $\rho \in S(\mathcal{H}_A)$ be a state. A purification for ρ is some pure state $\rho' \in S(\mathcal{H}_A \otimes \mathcal{H}_B)$ where $\mathcal{H}_B$ is an ancillary Hilbert space such that $\rho = \text{Tr}_{\mathcal{H}_B}(\rho')$.

Purification

Lemma (Canonical Purification)

Define $|\Omega\rangle := \sum_{i=0}^d |i\rangle \otimes |i\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ with $\mathcal{H}_A \cong \mathcal{H}_B$, and let $\rho \in S(\mathcal{H}_A)$. We define

$$\Psi_\rho := (\sqrt{\rho} \otimes I)|\Omega\rangle\langle\Omega|(\sqrt{\rho} \otimes I) \in S(\mathcal{H}_A \otimes \mathcal{H}_B) \quad (7)$$

is a purification for ρ .

Lemma (Uniqueness up to Unitary)

Let $\rho \in S(\mathcal{H}_A)$, and let $\rho' \in S(\mathcal{H}_A \otimes \mathcal{H}_B)$ be a purification. Then $\rho'' \in S(\mathcal{H}_A \otimes \mathcal{H}_B)$ is also a purification if and only if there exists a unitary $U \in L(\mathcal{H}_A \otimes \mathcal{H}_B)$ such that $\rho'' = (I_A \otimes U_B)\rho'(I_A \otimes U_B^\dagger)$.

Lemma (r-dimensional ancillary necessary and sufficient)

Let $\rho \in S(\mathcal{H}_A)$ with $\dim \mathcal{H} = d$ and $\text{rank } \rho = r \leq d$. Then, for all Hilbert spaces $\mathcal{H}_B$ with $\dim \mathcal{H}_B \geq r$, ρ has a purification in $S(\mathcal{H}_A \otimes \mathcal{H}_B)$

Haar Measure

Definition (Haar measure on $U(d)$)

There exists a unique probability measure called the Haar measure μ_H over the unitary group $U(d)$ that is left and right invariant on $U(d)$.

For all integrable functions $f : U(d) \rightarrow \mathcal{H}$ and unitaries $V \in U(d)$, the Haar measure allows us to take expectation values:

$$\mathbb{E}_{U \sim \mu_H} [f(u)] := \int_{U(d)} f(U) d\mu_H(U) \quad (8)$$

The invariance then looks like:

$$\mathbb{E}_{U \sim \mu_H} [f(U)] = \mathbb{E}_{U \sim \mu_H} [f(VU)] = \mathbb{E}_{U \sim \mu_H} [f(UV)] \quad (9)$$

Random Purification Channel

Theorem (Existence and Uniqueness of Random Purification)

Let $\mathcal{H}_A \cong \mathcal{H}_B$. There exists a unique channel $\mathcal{E} : L(\mathcal{H}_A)^{\otimes n} \rightarrow L(\mathcal{H}_A)^{\otimes n} \otimes L(\mathcal{H}_B)^{\otimes n}$ such that for all $\rho \in S(\mathcal{H}_A)$:

1. $\mathcal{E}(\rho^{\otimes n}) = \mathbb{E}_{U \sim \mu_H} [(I \otimes U^{\otimes n}) \Psi_{\rho^{\otimes n}} (I \otimes U^{\dagger \otimes n})]$
2. $\mathcal{E} \circ \text{Sym} = \mathcal{E}$

In particular, this map is given by

$$\mathcal{E}(\cdot) = \sqrt{R_n}(\cdot \otimes I)\sqrt{R_n} \quad (10)$$

where

$$R_n := \mathbb{E}_{U \sim \mu_H} [(I \otimes U^{\otimes n}) |\Omega\rangle\langle\Omega|^{\otimes n} (I \otimes U^{\dagger \otimes n})] \quad (11)$$

Random Purification Channel Cont.

Lemma

For all states $\rho_A \in S(\mathcal{H}_A)$, we have:

$$(\sqrt{\rho_A^{\otimes n}} \otimes I_B) R_n (\sqrt{\rho_A^{\otimes n}} \otimes I_B) = \sqrt{R_n} (\rho_A^{\otimes n} \otimes I_B) \sqrt{R_n} \quad (12)$$

This lemma can now be used to show that the channel gives the correct expectation value. For a state $\rho_A \in S(\mathcal{H}_A)$:

$$\mathcal{E}(\rho^{\otimes n}) = \sqrt{R_n} (\rho_A^{\otimes n} \otimes I_B) \sqrt{R_n} \quad (13)$$

$$= (\sqrt{\rho_A^{\otimes n}} \otimes I_B) R_n (\sqrt{\rho_A^{\otimes n}} \otimes I_B) \quad (14)$$

$$= \mathbb{E}_{\mathcal{U}_B} [(\sqrt{\rho_A^{\otimes n}} \otimes I_B) (I_A \otimes U_B^{\otimes n}) |\Omega\rangle \langle \Omega| (I_A \otimes U_B^{\dagger \otimes n}) (\sqrt{\rho_A^{\otimes n}} \otimes I_B)] \quad (15)$$

$$= \mathbb{E}_{\mathcal{U}_B} [(I_A \otimes U_B^{\otimes n}) \Psi_\rho (I_A \otimes U_B^{\dagger \otimes n})] \quad (16)$$

Lemma

$$\text{Sym}_n(L(\mathcal{H}_A^{\otimes n})) = \text{Span}\{\rho^{\otimes n} : \rho \in S(\mathcal{H}_A)\} \quad (17)$$

Uniqueness of the Random Purification Channel $\mathcal{E}$.

We can write $L(\mathcal{H}_A^{\otimes n})$ into the direct sum of its symmetric subspace and its complementary subspace:

$$L(\mathcal{H}_A^{\otimes n}) = \text{Sym}_n(L(\mathcal{H}_A^{\otimes n})) \oplus \text{Sym}_n(L(\mathcal{H}_A^{\otimes n}))^\perp \quad (18)$$

Take some $X \in L(\mathcal{H}_A^{\otimes n})$, and write $X = X_{\text{Sym}} + X_\perp$. Sym is the orthogonal projection onto the symmetric subspace, so $\text{Sym}_n(X) = X_{\text{Sym}}$. Since $\mathcal{E} \circ \text{Sym}_n = \mathcal{E}$, then $\mathcal{E}(X) = \mathcal{E}(\text{Sym}_n(X)) = \mathcal{E}(X_{\text{Sym}})$. Now, choose a basis for $\text{Sym}_n(L(\mathcal{H}_A^{\otimes n}))$ of the form $\{\rho^{\otimes n}\}$. Property (1) specifies $\mathcal{E}$ on every $\rho^{\otimes n}$, and thus on every basis element. Since linear maps are uniquely determined by their action on a basis, then $\mathcal{E}$ is unique. $\square$

Open Problem

"A natural open problem is to find an equally simple construction of the random purification channel tailored to rank- r inputs, in which the purifying system can also be taken to have dimension r " - Girardi et al. (2026)

Current state of the problem

From here out, let $\mathcal{H}_A, \mathcal{H}_B$ be Hilbert spaces with $\dim(\mathcal{H}_A) = d$ and $\dim(\mathcal{H}_B) = r$.

Theorem (Existence of Partial Random Purification)

There exists a channel $\mathcal{E} : L(\mathcal{H}_A)^{\otimes n} \rightarrow L(\mathcal{H}_A)^{\otimes n} \otimes L(\mathcal{H}_B)^{\otimes n}$ such that for all $\rho \in S(\mathcal{H}_A)$ with $\text{rank}(\rho) \leq r$:

1. $\mathcal{E}(\rho^{\otimes n}) = \mathbb{E}_{U \sim \mu_H} \left[(I \otimes U^{\otimes n}) \Psi_{\rho^{\otimes n}}^r (I \otimes U^{\dagger \otimes n}) \right]$
2. $\mathcal{E} \circ \text{Sym} = \mathcal{E}$

Here $\Psi_{\rho^{\otimes n}}^r$ is an arbitrary purification of $\rho^{\otimes n} \in \mathcal{H}_A \otimes \mathcal{H}_B$.

Lemma (No Partial Cup)

Let $d \geq 2$ and $r < d$. Then, for all $|\Omega_r\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ there exists $\rho \in S(\mathcal{H}_A)$ with $\text{rank } \rho \leq r$ such that

$$\rho \neq \text{Tr}_B((\sqrt{\rho} \otimes I)|\Omega_r\rangle\langle\Omega_r|(\sqrt{\rho} \otimes I)) \quad (19)$$

Proof.

Pick some $|\Omega_r\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$. We can write the Schmidt decomposition

$$|\Omega\rangle_r = \sum_{1 \leq i \leq r} \lambda_i |\psi_i\rangle |\phi_i\rangle \quad (20)$$

where $\lambda_i \in \mathbb{R}$ and $|\psi_i\rangle, |\phi_i\rangle$ are orthonormal bases for $\mathcal{H}_A$ and $\mathcal{H}_B$ respectively. Since $r < d$ we can pick a $|\psi_{r+1}\rangle$ orthonormal to all $|\psi_i\rangle$. Define $\rho' = 1/2|\psi_{r+1}\rangle\langle\psi_{r+1}| + 1/2|\psi_1\rangle\langle\psi_1|$. Then we have

$$\text{Tr}_B((\rho' \otimes I)|\Omega_r\rangle\langle\Omega_r|) = \sum_{i,j} \text{Tr}_B(\lambda_i \lambda_j |\psi_i\rangle\langle\psi_j| \rho' \otimes |\phi_i\rangle\langle\phi_j|) \quad (21)$$

$$= \sum_i \lambda_i^2 |\psi_i\rangle\langle\psi_i| \rho' \quad (22)$$

$$= \frac{\lambda_1^2}{2} |\psi_1\rangle\langle\psi_1| \quad (23)$$

$$\neq \rho' \quad (24)$$

Theorem (Non-Uniqueness of Partial Random Purification)

For all $n, r, d \in \mathbb{N}$ with $r < \left(\frac{dn}{e \cdot (d^2 + n - 1)} \right)$ there are multiple channels $\mathcal{E} : L(\mathcal{H}_A)^{\otimes n} \rightarrow L(\mathcal{H}_A)^{\otimes n} \otimes L(\mathcal{H}_B)^{\otimes n}$ such that for all $\rho \in S(\mathcal{H}_A)$ with $\text{rank } \rho \leq r$:

1. $\mathcal{E}(\rho^{\otimes n}) = \mathbb{E}_{U \sim \mu_H} \left[(I \otimes U^{\otimes n}) \Psi_{\rho^{\otimes n}}^r (I \otimes U^{\dagger \otimes n}) \right]$
2. $\mathcal{E} \circ \text{Sym} = \mathcal{E}$

Here $\Psi_{\rho^{\otimes n}}^r$ is an arbitrary purification of $\rho^{\otimes n} \in \mathcal{H}_A \otimes \mathcal{H}_B$.

Lemma

Let $n, r, d \in \mathbb{N}$. For $r < \left(\frac{dn}{e(d^2+n-1)} \right)$,

$$R := \text{Span}\{\rho^{\otimes n} : \rho \in S(\mathcal{H}_A), \text{rank } \rho \leq r\} \subsetneq \text{Sym}_n(L(\mathcal{H}_A^{\otimes n})) \quad (25)$$

Proof of Lemma

1. Since clearly $I \in \text{Sym}_n(L(\mathcal{H}_A^{\otimes n}))$ we will show that I is not in the LHS.
2. Pick a basis for the LHS $\mathcal{B} = \{\rho_i^{\otimes n}\}$ where each $\rho_i \in S(\mathcal{H}_A)$ with $\text{rank } \rho_i \leq r$.
3. Observe that $|\mathcal{B}| \leq \binom{d^2+n-1}{n}$
4. Assume for contradiction that for some $c_i \in \mathbb{C}$

$$\sum c_i \rho_i^{\otimes n} = I \quad (26)$$

5. Applying rank to LHS

$$\text{rank} \left(\sum c_i \rho_i^{\otimes n} \right) \leq \sum \text{rank}(c_i \rho_i^{\otimes n}) \leq r^n \binom{d^2+n-1}{n} \quad (27)$$

6. Algebra gives

$$r < \left(\frac{dn}{e(d^2+n-1)} \right) \implies r^n \binom{d^2+n-1}{n} < d^n \quad (28)$$

and so the ranks of both sides of equation (26) cannot be equal.

Proof of Non-Uniqueness of Partial Random Purification

1. Show that $L(\mathcal{H}^{\otimes n}) = R \oplus F \oplus \text{Sym}_n(\mathcal{H})^\perp$ for some non-trivial subspace $F \subseteq L(\mathcal{H}^{\otimes n})$
2. Show that the direct sum of channels is a channel.
3. Pick two channels on F .

Thank You

We thank Dr. Eugene Tang and Samuel Dai for their mentorship through this process, as well as Dr. Anadil Saeed Rao for his valuable feedback and representation theory expertise. We acknowledge financial support from the Northeastern University math department.

Girardi, F., Mele, F. A., and Lami, L. (2026). Random purification channel made simple.